

DataSight PRO

Streamlining Data Analysis & ML Preparation

An Interactive Tool for Exploration, Visualization, and Preprocessing

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INEW 2332 - Comprehensive Software Project



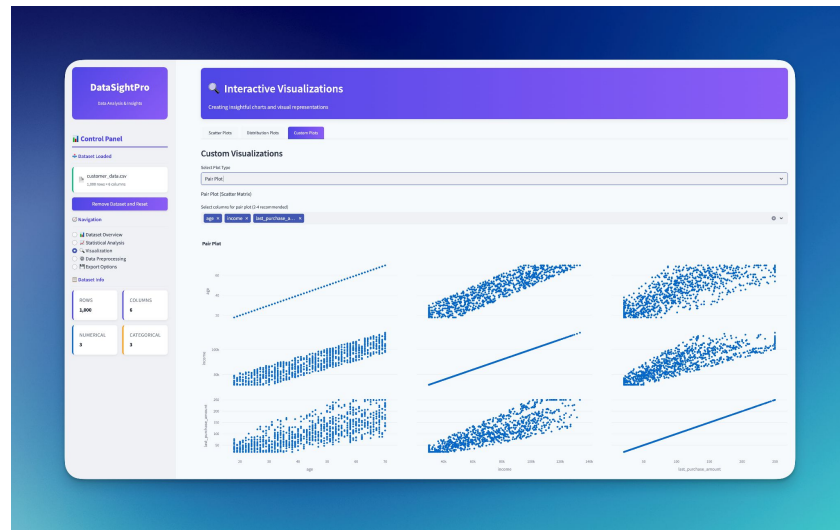
The Challenge & The Solution

Challenge: Data analysis and preparation for Machine Learning can be complex, time-consuming, and require significant coding expertise.

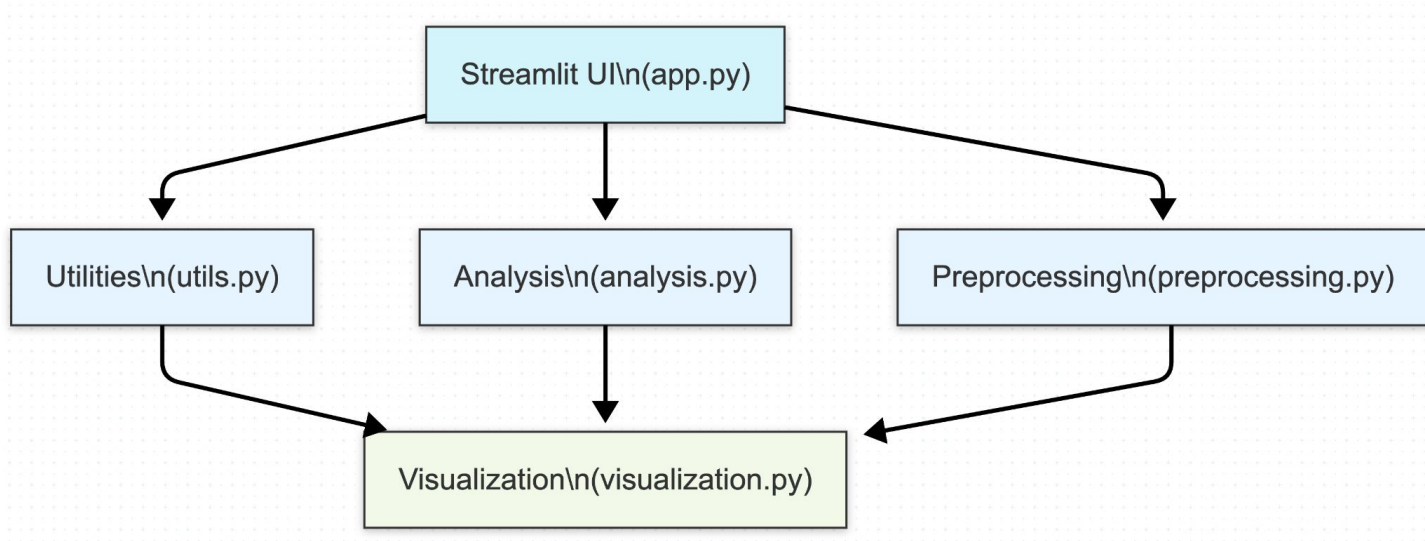
Solution: DataSight PRO - An interactive web application built with Python & Streamlit.

Goal: To streamline the data analysis workflow by providing an intuitive interface for exploration, statistical analysis, visualization, and preprocessing – making it accessible without extensive programming.

Target Audience: Data Scientists, Analysts, ML Engineers, Researchers, Students, Anyone working with data.



Architecture Overview



The Streamlit UI layer interacts with distinct backend modules -Utilities, Analysis, Preprocessing, and Visualization.
The main application coordinates data flow between these modules..

Intuitive UI

DataSightPro

Data Analysis & Insights

Control Panel

Upload Dataset

Upload a CSV, Excel, or JSON file to begin analysis

Drag and drop file here

Limit: 200MB per file • CSV, XLSX, JSON

Browse files

Welcome to DataSightPro

A comprehensive platform for data analysis and machine learning preparation

Data Exploration

Quickly understand your dataset with automatic profiling, data type detection, and missing value analysis. Gain insights into your data's structure and quality.

Advanced Analytics

Perform statistical analysis, correlation studies, and distribution examinations with beautiful visualizations and detailed metrics.

Interactive Visualization

Create beautiful, interactive charts with Plotly to uncover patterns and relationships in your data. Customize visualizations for your specific needs.

ML Preparation

Preprocess your data with encoding, scaling, and feature engineering tools designed for machine learning workflows. Export processed data in multiple formats.

Getting Started

Follow These Steps

- Upload a dataset**
Use the file uploader in the sidebar to load your CSV, Excel, or JSON file.
- Explore dataset statistics**
View data summary, distributions, and correlations in the Dataset Overview.
- Preprocess your data**
Clean, transform, and prepare your data for machine learning models.
- Export your results**
Download your processed data in various formats for further use.

Supported File Formats

Compatible Formats

- CSV (.csv)**
Import and export comma-separated values files
- Excel (.xlsx)**
Import and export Microsoft Excel spreadsheets
- JSON (.json)**
Import and export JavaScript Object Notation files

Try a Sample Dataset

Customer Data Sample

Try out DataSightPro's features with our prepared customer dataset.

Load Sample Dataset

DataSight PRO Workflow



Follow These Steps

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Uncover Insights Visually

Control Panel

Dataset Loaded

customer_data.csv
1,000 rows x 6 columns

Remove Dataset and Reset

Navigation

- ☐ Dataset Overview
- ☐ Statistical Analysis
- ☒ Visualization
- ☐ Data Preprocessing
- ☐ Export Options

Dataset Info

ROWS

1,000

COLUMNS

6

NUMERICAL

3

CATEGORICAL

3

Scatter Plots

Distribution Plots

Custom Plots

Custom Visualizations

Select Plot Type

3D Scatter Plot

3D Scatter Plot

X-axis

income



Y-axis

age

Z-axis

last_purchase_amount

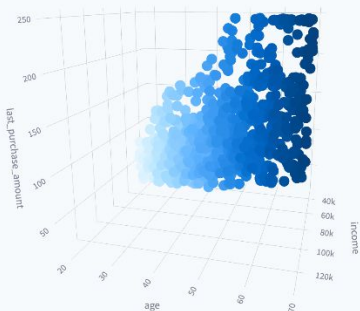


Color points by (optional)

age



3D Scatter Plot: income vs age vs last_purchase_amount, colored by age



Prepare Data for Machine Learning

Handle Missing Values: Impute using Mean, Median, Mode, or Constant Value

Encode Categorical Data: Convert text to numbers using One-Hot or Label Encoding

Normalize Data: Normalize data using MinMax, Z-Score, or Robust Scaling (handles outliers)

Feature Selection: Select variables manually or by correlation

Data Preprocessing

Prepare your dataset for machine learning and further analysis

Missing Value Handling

Categorical Encoding

Feature Scaling

Feature Selection

Missing Value Handling

Columns with missing values:

- income: 0.30% missing
- education: 0.30% missing

Select handling strategy

Fill missing values with mean/mode

Apply - Fill with Mean/Mode

Missing Value Handling

Categorical Encoding

Feature Scaling

Feature Selection

Feature Selection

Select feature selection method:

Method

☒ Correlation-based Selection
☐ Manual Selection

Correlation threshold

0.10 0.80 1.00

Highly correlated feature pairs:

- age & income: 0.9081
- income & last_purchase_amount: 0.8398

Suggested features to remove (highest correlation counts):

- income: appears in 2 high correlation pairs
- age: appears in 1 high correlation pairs
- last_purchase_amount: appears in 1 high correlation pairs

Select features to remove

income x

Remove Selected Features



Instant Insights, Endless Evolution

Core Benefits:

- **Speed:** Saves time and delivers instant results
- **Simplicity:** Reduces complexity of data analysis via code-minimal interface
- **Accessibility:** Empowers users with diverse skill levels

Future Enhancements:

- AutoML Integration
- Advanced Visualizations
- Direct Database Connectors
- Guided Workflows & Customizable Dashboards



DataSight PRO

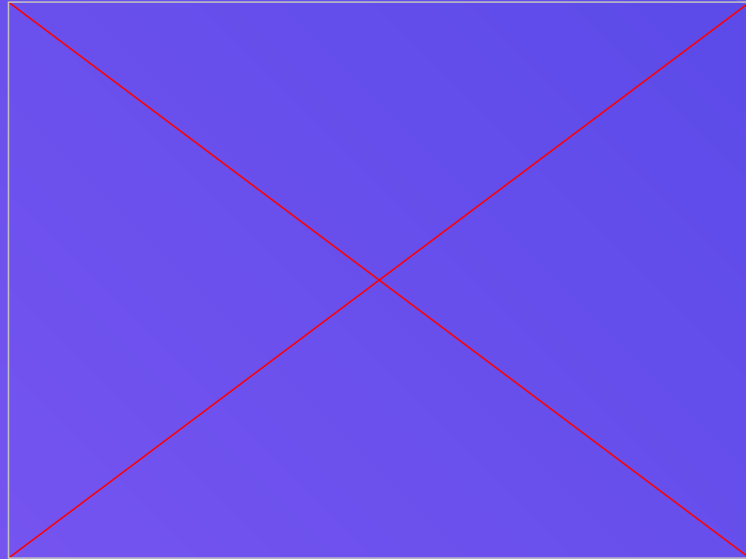
Empowering Insights for
Everyone



Enjoy your Data!



GitHub Repo



Project Demo